Welcome

- Welcome to today's presentation
- Topic for today: Containerization (Docker Basics)
- We will understand containers, Docker, and how they differ from Virtual Machines
- Focus on practical cloud and DevOps relevance

Agenda

- Introduction to containerization
- Understanding Virtual Machines
- Containers vs Virtual Machines
- Introduction to Docker
- Docker architecture and components
- Real world applications and conclusion

What is Containerization?

- Containerization is a method of packaging applications with all dependencies.
- It ensures software runs the same in different computing environments.
- Containers isolate applications from each other.
- They are lightweight compared to traditional virtualization.

Why Containerization Matters

- Improves portability across operating systems.
- Speeds up software deployment and testing.
- Reduces conflicts between application dependencies.
- Commonly used in cloud computing and microservices architecture.

Virtual Machines (VMs)

- Virtual Machines emulate full hardware systems.
- Each VM runs its own operating system.
- VMs run on a hypervisor layer.
- They provide strong isolation but consume more memory and CPU.

Containers Explained

- Containers share the host operating system kernel.
- They only include application code and dependencies.
- They start quickly and use fewer resources.
- This makes them ideal for scalable cloud applications.

VM vs Containers

- VMs require full operating systems.
- Containers share the host OS.
- VMs are heavier and slower to start.
- Containers are lightweight and deploy quickly.
- Containers are commonly used in DevOps pipelines.

Introduction to Docker

- Docker is the most widely used container platform.
- It allows developers to build, ship, and run containers easily.
- Docker simplifies environment management.
- It plays a key role in modern DevOps workflows.

Docker Architecture

- Docker Engine runs and manages containers.
- Docker Images are templates used to create containers.
- Docker Containers are running instances of images.
- Docker Hub is a repository for sharing container images.

Conclusion

- Containerization improves efficiency and portability.
- Docker makes container management simple.
- Containers are essential for cloud-native applications.
- Understanding containers is crucial for modern software development.